

STRANGER THINGS: THE DEMOGORGON BOSS FIGHT

A 2D RPG Boss Fight Game Built in Java Swing

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ABSTRACT

This project presents the development of a 2D RPG Boss Fight game inspired by Stranger Things using Java Swing. The game demonstrates real-time gameplay, collision detection, enemy artificial intelligence, animated visual effects and sound integration. The project uses object-oriented programming concepts and a thread-based game loop to create an interactive desktop game without external game engines.

CHAPTER 1: INTRODUCTION

1.1 Background

This project recreates the Demogorgon battle in a 2D RPG format inspired by Stranger Things. It demonstrates game development concepts using Java Swing.

1.2 Problem Statement

Most beginner Java projects are console-based and lack graphical interaction. This project solves that by building a real-time playable desktop game.

1.3 Objectives

Develop a playable game, implement enemy AI, integrate sound, and demonstrate object-oriented programming.

1.4 Scope

Includes player movement, attacks, enemy AI, intro screens, health bars and game over conditions.

CHAPTER 2: LITERATURE REVIEW

2D RPG systems commonly use sprite rendering, collision detection and finite state machines. Java Swing supports Graphics2D rendering and event handling suitable for game development.

CHAPTER 3: SYSTEM DESIGN

The system follows modular OOP design.

Class	Purpose
Main	Starts application
GamePanel	Game loop and rendering
Player	Player movement and attacks
Enemy	Boss AI
KeyHandler	Keyboard controls
SoundManager	Audio
IntroPanel	Intro animation

CHAPTER 4: IMPLEMENTATION

The game uses Java Swing and AWT Graphics2D. A separate thread runs the update loop. Collision detection uses Rectangle intersection. The boss follows roaming, warning, charging and recovering states.

CHAPTER 5: OUTPUT

Output includes intro screens, gameplay visuals, health bars, and victory or defeat screens.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

The project successfully demonstrates Java game development. Future improvements may include multiplayer, more bosses and score systems.

REFERENCES

1. Oracle Java Swing Documentation
2. Java AWT API
3. Finite State Machine Concepts